

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
7	Indicative content In systemic circulation: • {High pressure/ pressure increase} (to 120 mmHg) in {aorta/ arteries} is due to ventricular systole/ contraction of left ventricle. • High pressure needed to pump blood {round body/ long distance} • Pressure decrease (to 80 mmHg) in {arteries / aorta} is due to {diastole/ ventricular relaxation}. • High pressure is maintained due to elastic recoil of artery walls / closure of {semilunar / aortic} valve. • Pressure decrease in {arterioles/ capillaries} is due to {frictional resistance / blood flowing through a greater (total) {cross sectional/ surface area}. • Pressure decrease in capillaries is also due to tissue fluid formation. • low pressure/ flow rate in capillaries necessary for diffusion • Blood flow in veins due to massaging effect of skeletal muscles. In pulmonary circulation: • Pressure increase in pulmonary arteries due to {ventricular systole/ contraction of thinner walled right ventricle}. • Lower pressure (than systemic) due to {shorter distance / (only) to lungs}. • Pressure decrease (to about 8 mmHg) in pulmonary arteries is due to diastole / ventricle relaxation. • Pressure in capillaries is lower than systemic to prevent tissue fluid formation in the alveoli / reduce damage to alveoli / allow time for gas exchange.	6	3		9		

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	Benefit over single circulation: • Correct description of single and double circulation • Pressure is lost when blood passes through (gill) capillaries. • so blood flow through the systemic circulation of a fish is slower. • Double circulation maintains high pressure (in systemic circulation) • to meet the high metabolic demands of a mammal.						

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	7-9 marks Detailed explanation of systemic circulation And Detailed explanation of pulmonary circulation And Detailed explanation of benefit over single circulation <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Any two from Explanation of systemic circulation Explanation of pulmonary circulation Explanation of benefit over single circulation <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i>						

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-3 marks Brief explanation of systemic circulation OR Brief explanation of pulmonary circulation OR Brief explanation of benefit over single circulation <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 6 total	6	3	0	9	0	0